hw2_writeup

February 4, 2026

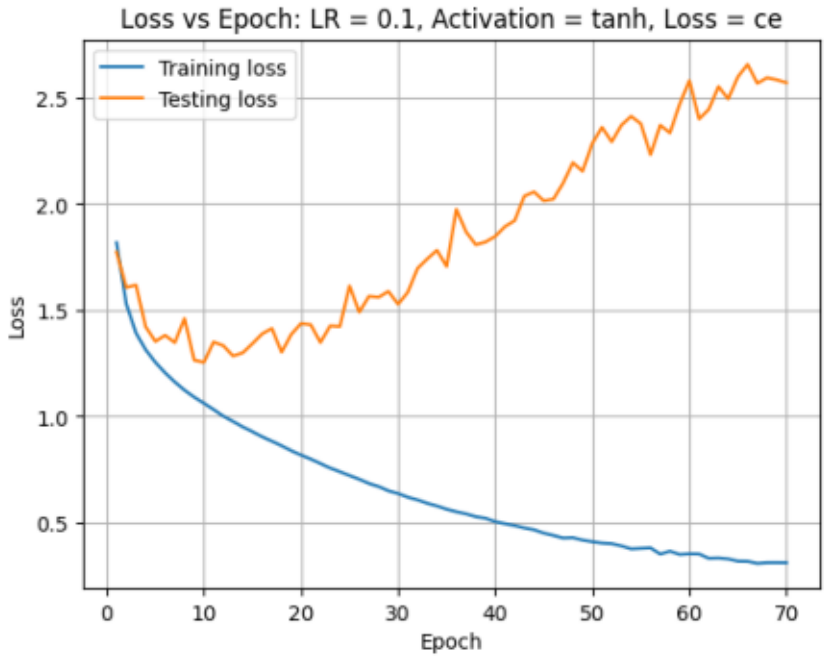
Lex Albrandt
CS440
HW 2 writeup

[1]:
```
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import numpy as np
import pandas as pd
```

0.1 Question 1

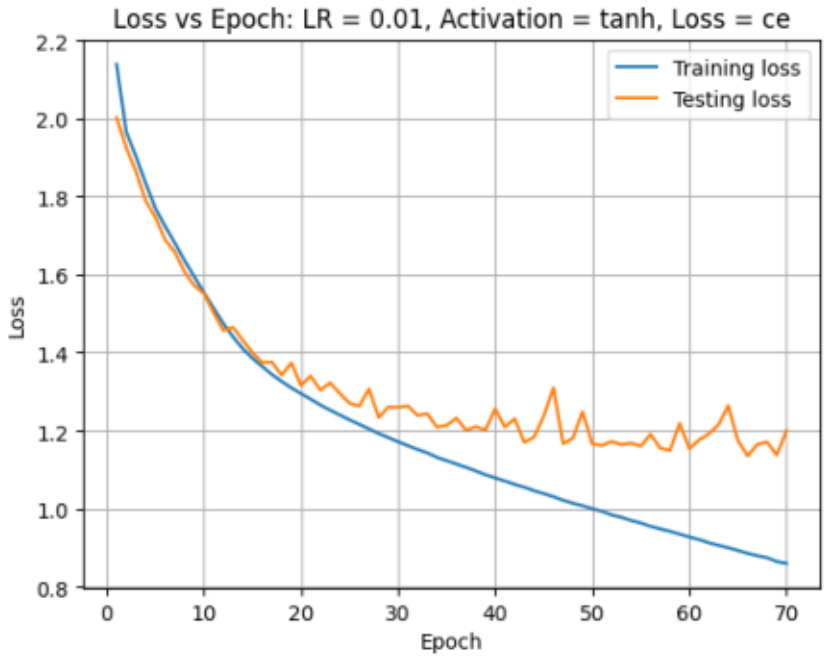
Note: All hyperparameters are indicated in the title of each figure for this section

[2]:
```
img = mpimg.imread("./figures/Figure_1_test1.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



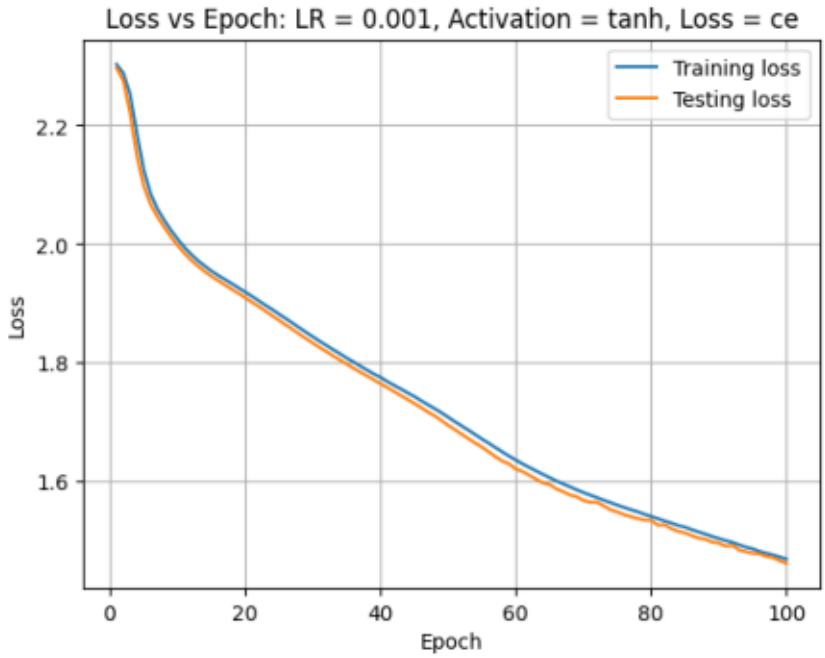
In this test we see that the training loss converges to a low value, indicating that it has learned the training data very well. We see, however, that the testing loss initially decreases with the training loss, but diverges and continues to increse. This indicates that the model has begun to overfit the training parameters and is no longer able to generalize, subsequently causing the model to perform poorly on unseen data. Because the learning rate in this test is quite high the gradient steps between epochs make much larger adjustments, which can cause the model to memorize the data, and lose the ability to generalize.

[3]:
```
img = mpimg.imread("./figures/Figure_1_test2.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



In this test, we see that the model does a much better job of learning patterns present in both the training and test data, however, after about 20 epochs, the model clearly begins to overfit the data, learning patterns like noise that do not exist in the test set.

[4]:
```
img = mpimg.imread("./figures/Figure_1_test3.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```

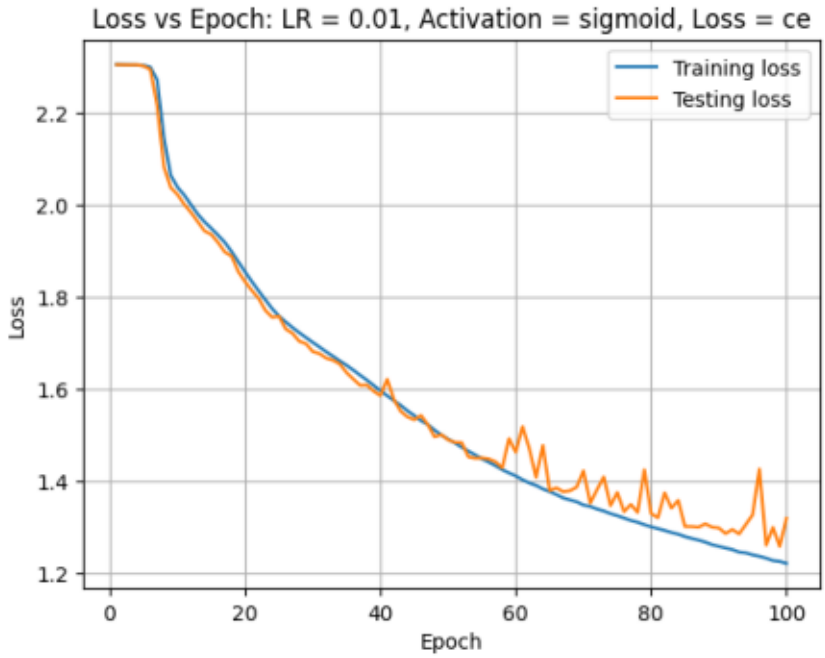


Here we see that the model training loss and testing loss are closely matched, indicating that the model is generalizing well, and applying learned patterns to unseen data, rather than memorizing the training set. The lower learning rate contributes to the model's stability, which prevents large shifts in the weights between epochs, and mitigates overfitting.

[5]:
```
img = mpimg.imread("./figures/Figure_1_test4.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```

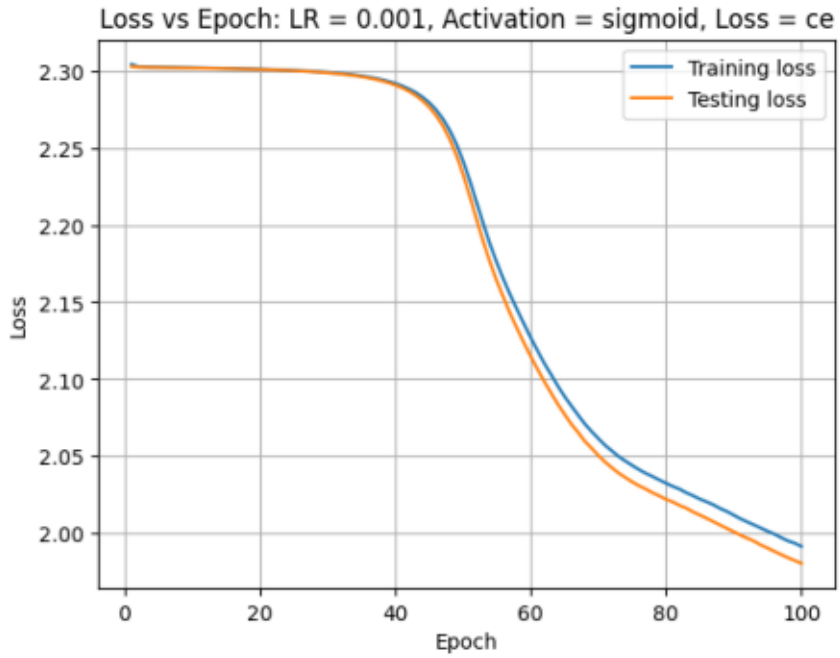
Once again, in this test we see that the learning rate causes an overfitting of the data and loss of generalization, despite a different activation function.

[6]:
```
img = mpimg.imread("./figures/Figure_1_test5.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



In this test we see that the training and test losses remain closely matched, but diverge after around 60 epochs, again indicating overfitting.

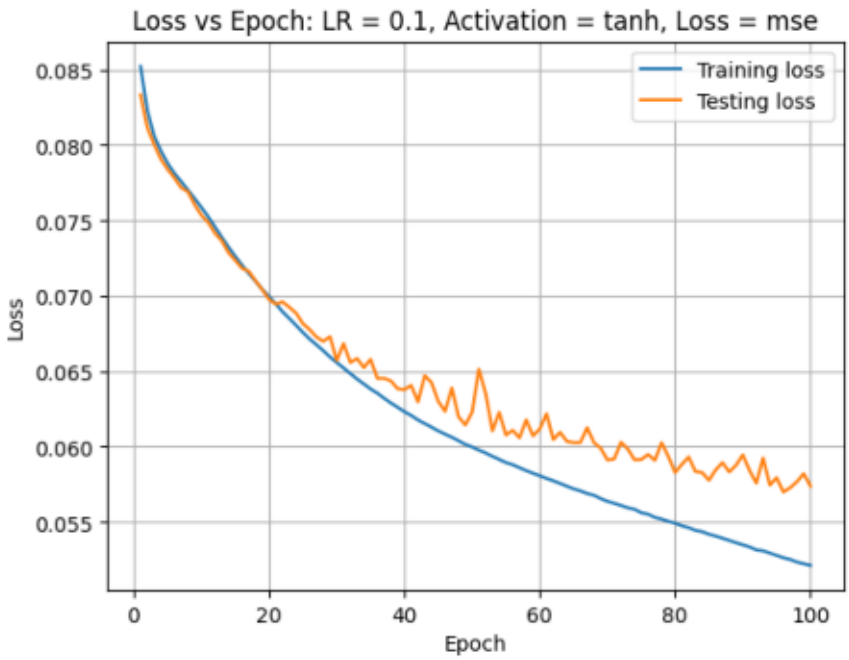
[7]:
```
img = mpimg.imread("./figures/Figure_1_test6.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



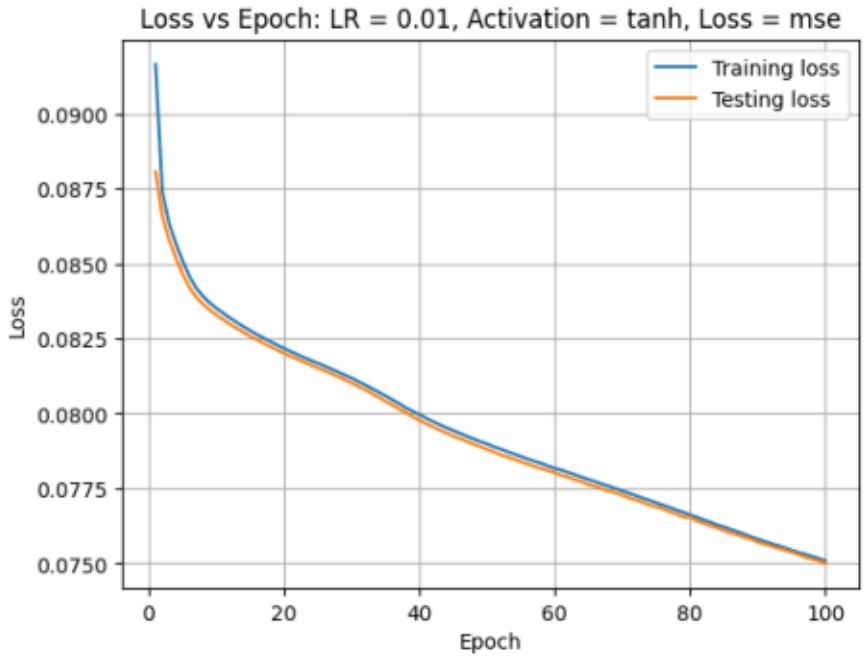
In ths test we again see closely matched training and test loss, and a pattern has started to emerge regarding performance: it appears that the most important determining factor for good generalization with this data and this model is a low learning rate of 0.001

For the sake of space I will show the next few test plots, as they follow a similar pattern to the plots shown above

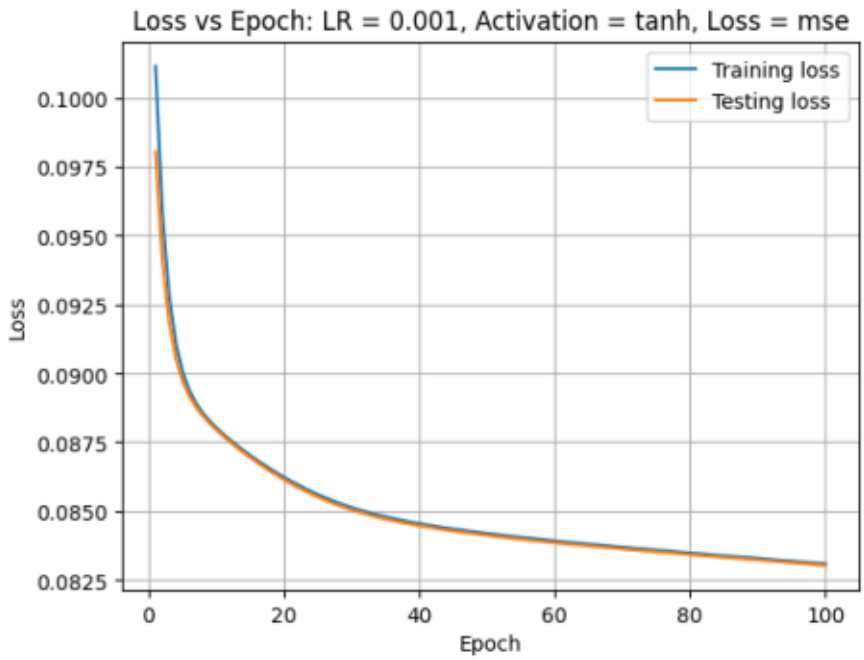
[8]:
```
img = mpimg.imread("./figures/Figure_1_test7.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



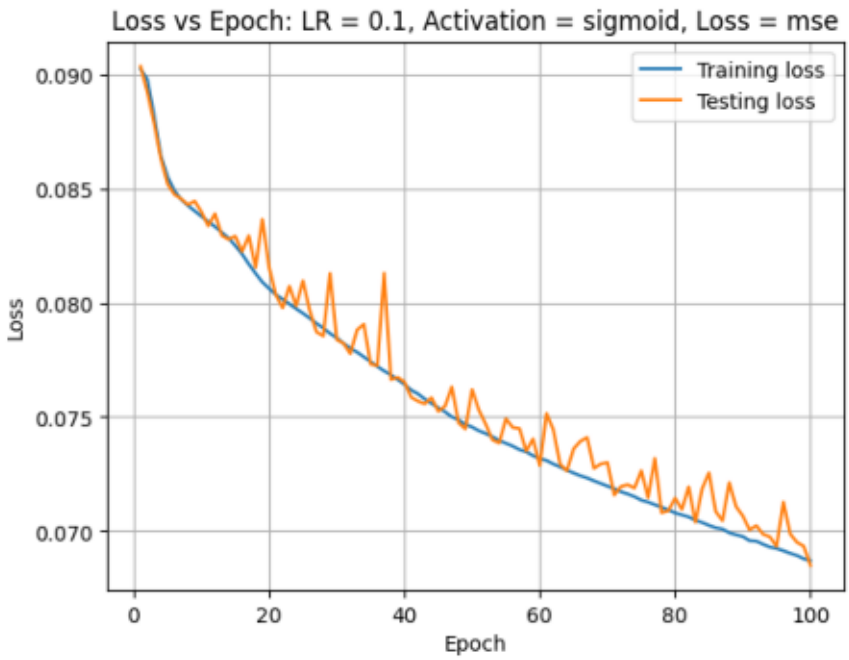
[9]:
```
img = mpimg.imread("./figures/Figure_1_test8.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



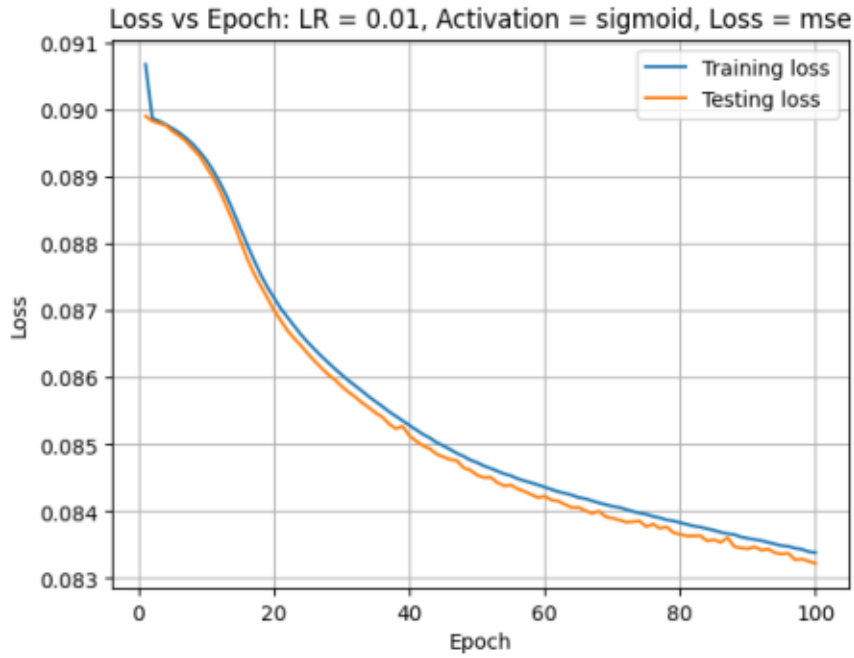
[10]:
```
img = mpimg.imread("./figures/Figure_1_test9.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



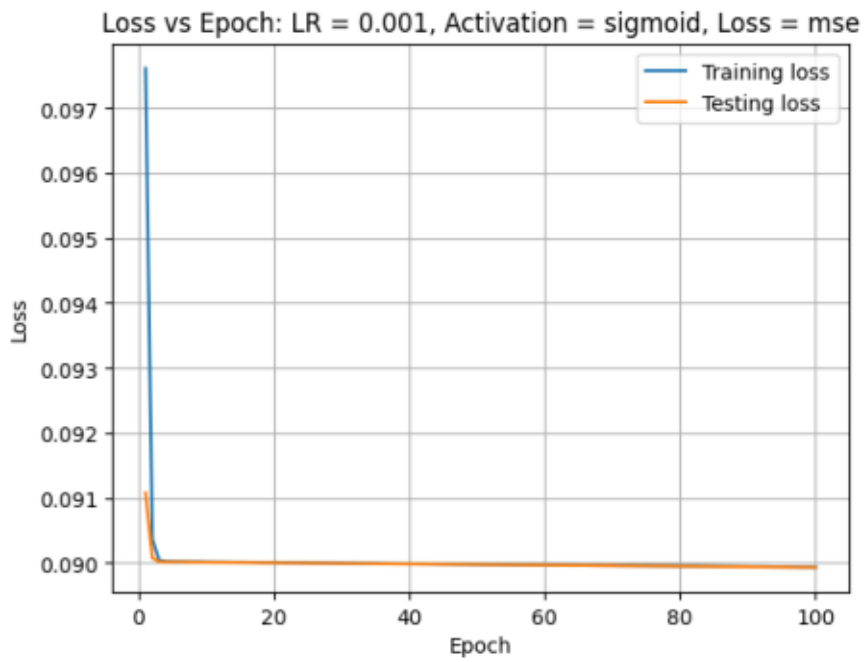
[11]:
```
img = mpimg.imread("./figures/Figure_1_test10.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



[12]:
```
img = mpimg.imread("./figures/Figure_1_test11.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



[13]:
```
img = mpimg.imread("./figures/Figure_1_test12.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```

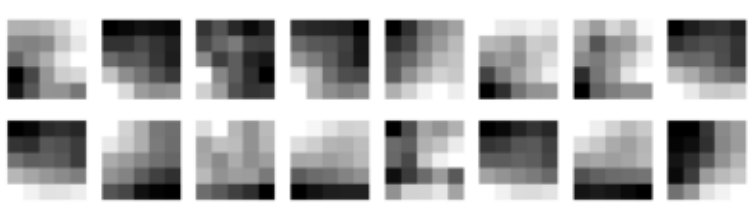


We can see, generally, that a lower learning rate increases the stability of the model, and allows it to generalize effectively.

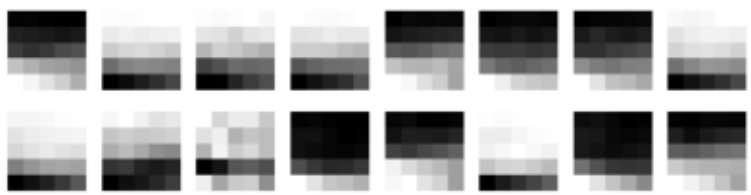
1 Feature Maps for Test 9

Below are three of the feature maps for test #9 which had the following hyperparamaters:
- Learning Rate: 0.001
- Activation Function: Tanh - Loss Function: MSE

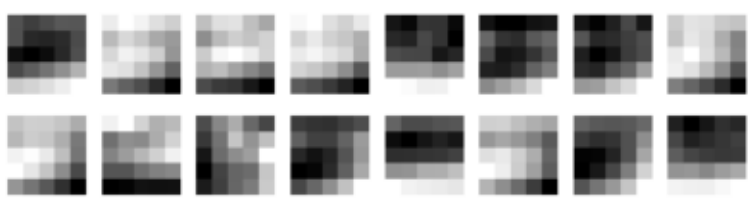
[14]:
```
img = mpimg.imread("./figures/fm_1.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



[15]:
```
img = mpimg.imread("./figures/fm_2.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



[16]:
```
img = mpimg.imread("./figures/fm_3.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



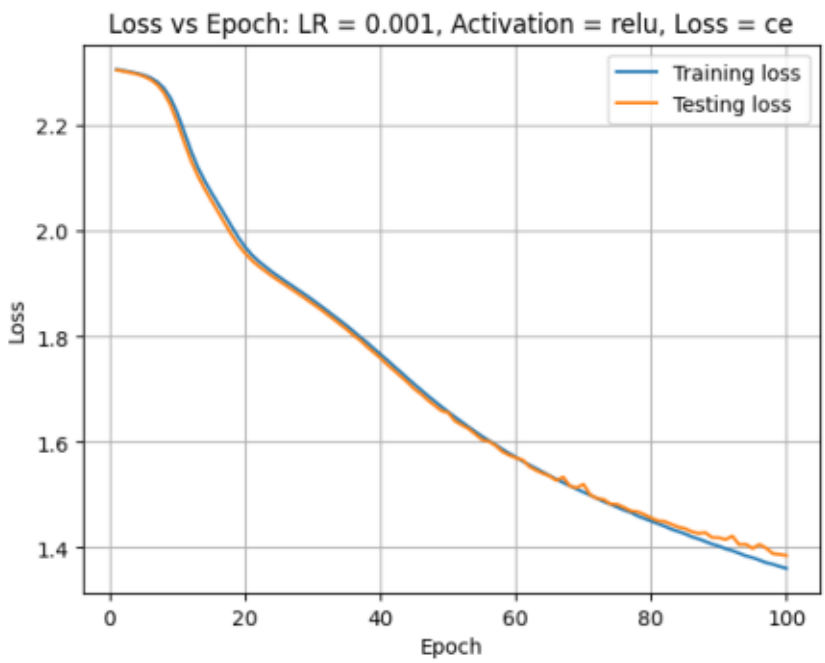
In the above feature maps we see that the feature maps at the last layer of convolution are abstract and generalized, indicating that the model is learning abstract lines or shapes to categorize images. The further we go into the model the more abstract these maps become, as we see in the maps above.

1.1 Question 2

Kernel size:
- 3x3

Hyperparameters for this test:
- Learning Rate: 0.001
- Activation Function: ReLU
- Loss Function: Cross Entropy

[17]:
```
img = mpimg.imread("./figures/relu_test.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



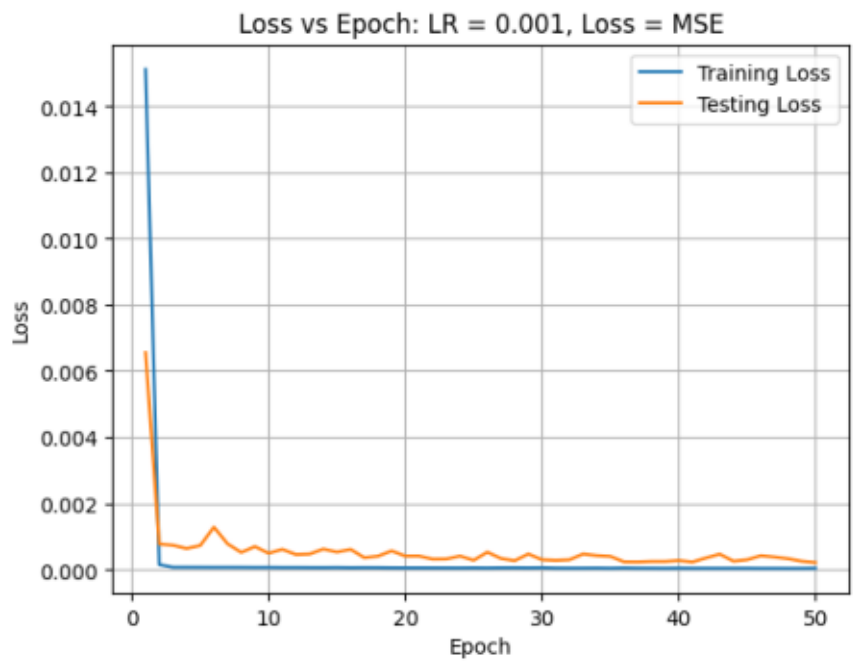
For this test we see a similar result to other tests that have a low learning rate: the model is generalizing well, and categorizing unseen data appropriately Because we are using the ReLU activation only neurons with positive output are "active" which reduces noise in the network, which in turn helps avoid overfitting and memorization and also improves generalization, which allows the model to perform better on unseen data.

1.2 Question 3

1.2.1 Stock Market Predictions

The task for this section was to create a RNN to predict the stock market prices for the 101st value after seeing 100 values of high prices from the Dow Jones data set. I chose to use an LSTM RNN for this section because it is good at capturing trends and cycles in stock prices. A typical RNN could also be used, but to my understanding would suffer from vanishing gradients, which would be a problem for long periods of time series data.

[18]:
```
img = mpimg.imread("./figures/rnn_stocks.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



We see above that the training loss decreases rapidly and approaches zero, which indicates that the model is closely fit to the training data. It's also worth noting that the testing loss is also low, and the model is generalizing fairly well. The testing loss remains low because predicting a stock value from the day before is highly autocorrelated, meaning the flucuation in value from day to day is miniscule, so the model does a good job of predicting values from day to day.

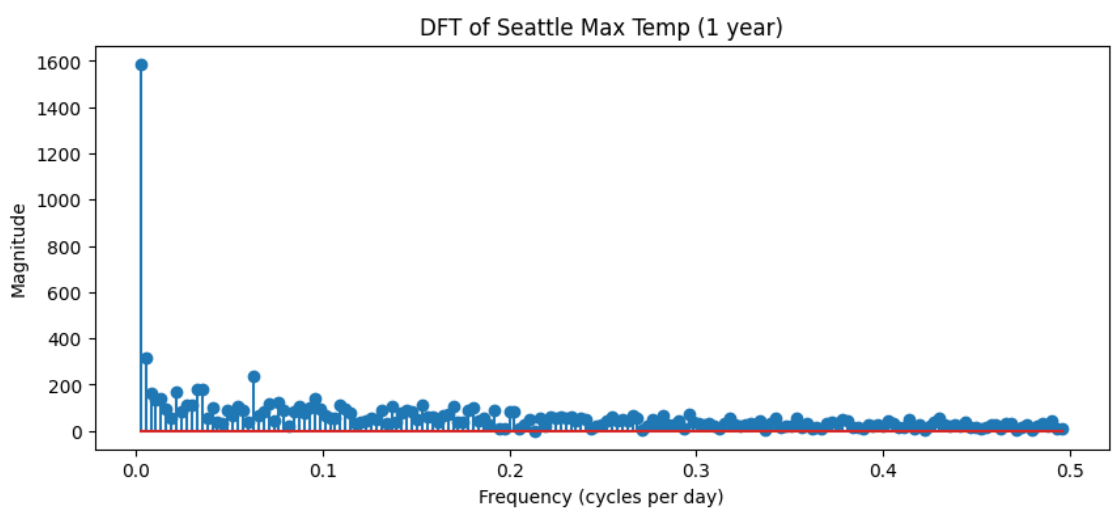
This is very different from predicting sin(x) because sin(x) is, by nature, periodic and smooth. This means the model must learn the amplitude and phase of the function to predict next values correctly, and must learn the pattern of the full function over time.
Because stocks have minute shifts day to day, short-term prediction is much easier.

1.2.2 Weather Data for Seattle

Discrete Fourier Transform

[19]:
```
df = pd.read_csv("seattle-weather.csv")

# Extract one year and three years of data
one_year_data = df["temp_max"].values[10:375]
```

```python
# Perform DFT
fft_vals = np.fft.fft(one_year_data)
fft_freqs = np.fft.fftfreq(len(one_year_data), d = 1)

# Compute magnitude spectrum
magnitude = np.abs(fft_vals)

skip = 1

# Plot data
plt.figure(figsize = (10, 4))
plt.stem(fft_freqs[skip:len(fft_freqs) // 2], magnitude[skip:len(magnitude) //␣
↪2])
plt.xlabel("Frequency (cycles per day)")
plt.ylabel('Magnitude')
plt.title("DFT of Seattle Max Temp (1 year)")
plt.show()

fft_list = list(zip(fft_freqs, magnitude))
fft_list_sorted = sorted(fft_list, key = lambda x: x[1], reverse = True)
for f, mag in fft_list_sorted[:10]:
    period_days = 1 / f if f != 0 else np.inf
    print(f"Frequency: {f:.4f} cycles/day, Magnitude: {mag:.2f}, period:␣
↪{period_days:.1f} days")
```

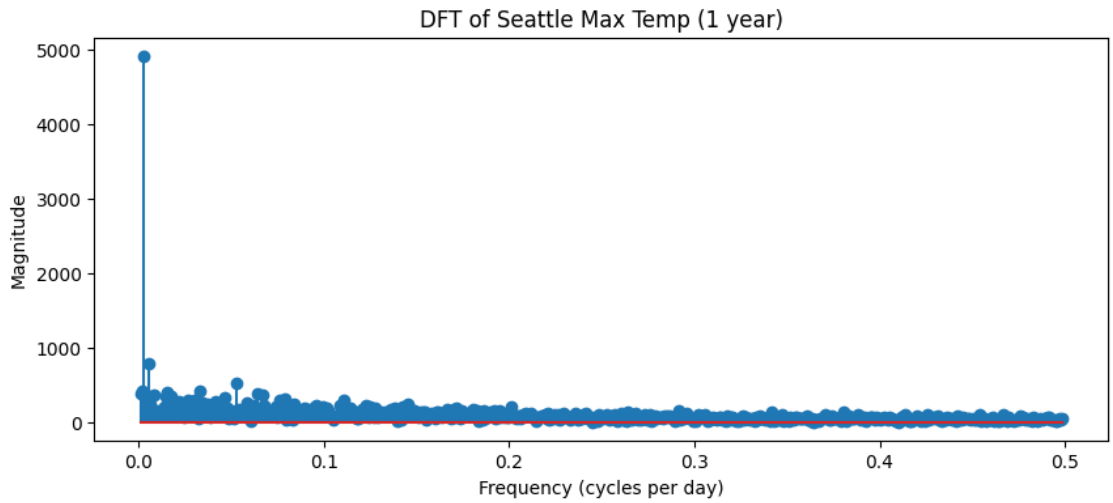



```
Frequency: 0.0000 cycles/day, Magnitude: 5571.40, period: inf days
Frequency: -0.0027 cycles/day, Magnitude: 1587.34, period: -365.0 days
Frequency: 0.0027 cycles/day, Magnitude: 1587.34, period: 365.0 days
Frequency: -0.0055 cycles/day, Magnitude: 317.99, period: -182.5 days
```

```
Frequency: 0.0055 cycles/day, Magnitude: 317.99, period: 182.5 days
Frequency: -0.0630 cycles/day, Magnitude: 236.18, period: -15.9 days
Frequency: 0.0630 cycles/day, Magnitude: 236.18, period: 15.9 days
Frequency: 0.0329 cycles/day, Magnitude: 182.23, period: 30.4 days
Frequency: -0.0329 cycles/day, Magnitude: 182.23, period: -30.4 days
Frequency: 0.0356 cycles/day, Magnitude: 177.18, period: 28.1 days
```

We can see on the DFT plot for one year of weather data there is a large frequency spike at around 0.002 cycles/day. To determine the period we use the formula:
Period $= \frac{1}{frequency} = \frac{1}{0.002} \approx 365$ days
This makes sense for weather cycles, given that the trends are often cyclical around the year. We also see spikes at 180 days, and 30 days, which again, makes sense given the cyclical nature of weather patterns.

[20]:
```python
df = pd.read_csv("seattle-weather.csv")

# Extract one year and three years of data
three_year_data = df["temp_max"].values[:1095]

# Perform DFT
fft_vals = np.fft.fft(three_year_data)
fft_freqs = np.fft.fftfreq(len(three_year_data), d = 1)

# Compute magnitude spectrum
magnitude = np.abs(fft_vals)

skip = 1

# Plot data
plt.figure(figsize = (10, 4))
plt.stem(fft_freqs[skip:len(fft_freqs) // 2], magnitude[skip:len(magnitude) //␣
↪2])
plt.xlabel("Frequency (cycles per day)")
plt.ylabel('Magnitude')
plt.title("DFT of Seattle Max Temp (1 year)")
plt.show()

fft_list = list(zip(fft_freqs, magnitude))
fft_list_sorted = sorted(fft_list, key = lambda x: x[1], reverse = True)
for f, mag in fft_list_sorted[:10]:
    period_days = 1 / f if f != 0 else np.inf
    print(f"Frequency: {f:.4f} cycles/day, Magnitude: {mag:.2f}, period:␣
↪{period_days:.1f} days")
```

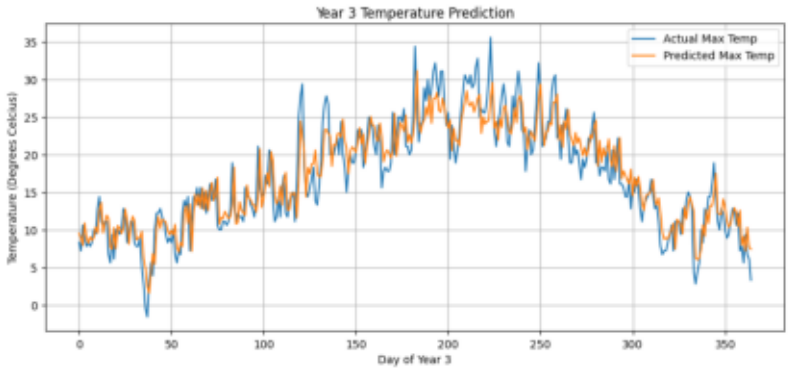



```
Frequency: 0.0000 cycles/day, Magnitude: 17653.00, period: inf days
Frequency: 0.0027 cycles/day, Magnitude: 4918.11, period: 365.0 days
Frequency: -0.0027 cycles/day, Magnitude: 4918.11, period: -365.0 days
Frequency: -0.0055 cycles/day, Magnitude: 781.68, period: -182.5 days
Frequency: 0.0055 cycles/day, Magnitude: 781.68, period: 182.5 days
Frequency: 0.0530 cycles/day, Magnitude: 514.10, period: 18.9 days
Frequency: -0.0530 cycles/day, Magnitude: 514.10, period: -18.9 days
Frequency: 0.0018 cycles/day, Magnitude: 418.75, period: 547.5 days
Frequency: -0.0018 cycles/day, Magnitude: 418.75, period: -547.5 days
Frequency: 0.0329 cycles/day, Magnitude: 416.60, period: 30.4 days
```

As with the one year data, we see spikes around 365 days, 180 days, and 30 days which makes sense given yearly and semi-yearly cycles of weather change.

3-year Weather prediction For this section of the assignment I used a very similar architecture to the stock
value prediction model, with the key difference of using a sliding-window approach with a look-back window of 10 dyas to predict the temperature for the next day. I used MSE and the Adam optimizer. I fed the true temperature values from the test set back into the sliding window at each timestep.

[21]:
```
img = mpimg.imread("./figures/temp.png")
plt.axis("off")
plt.imshow(img)
plt.show()
```



We observce that the predicted temperatures closely track the overall trend of actual temperatures. It is apparent that the trends appear smoother with less day-to-day variability. This is because the model was trained to minimize MSE over overlapping sliding windows, which causes the model to dampen the sharp fluctuations we see in the actual temperatures. We see that the model has generalized well, and has mitigated the tradeoffs between bias and variance. The model tends towards a lower variance because of averaging the patterns of temperature patterns but also has a higher bias as seen in the rapid changes of predictions. However, the model does balance the bias and variance and results in overall adequate generalization.

1.2.3 Sources Used

- https://www.kaggle.com/code/blurredmachine/lenet-architecture-a-complete-guide
- https://docs.pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html
- https://medium.com/thedeephub/rnns-in-action-predicting-stock-prices-with-recurrent-neural-networks-9155a33c4c3b